

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve the equation $|2x - 3| = 7$.

- (A) $x = 2$ or $x = 5$
- (B) $x = -2$ or $x = -5$
- (C) $x = 2$ or $x = -5$
- (D) $x = 5$ or $x = -2$
- (E) $x = 0$

Q3. Solve the equation $|x + 2| = 4$.

- (A) $x = -2$ or $x = 2$
- (B) $x = -6$ or $x = 2$
- (C) $x = -2$ or $x = 6$
- (D) $x = 2$ or $x = -6$
- (E) $x = 0$

Q4. Solve the equation $|3x - 1| = 11$.

- (A) $x = 2$ or $x = 4$
- (B) $x = -4$ or $x = 2$
- (C) $x = 4$ or $x = -4$
- (D) $x = 2$ or $x = -2$
- (E) $x = 0$

Q5. Solve the equation $|x - 5| = 2$.

- (A) $x = 7$ or $x = 3$
- (B) $x = 3$ or $x = 7$
- (C) $x = 5$ or $x = 0$
- (D) $x = 2$ or $x = 8$
- (E) $x = 0$

Q6. Solve the equation $|2x + 1| = 9$.

- (A) $x = 2$ or $x = -5$
- (B) $x = -2$ or $x = 5$
- (C) $x = 4$ or $x = -5$
- (D) $x = -4$ or $x = 2$
- (E) $x = 0$

Q7. Solve the equation $|x + 4| = 1$.

- (A) $x = -5$ or $x = -3$
- (B) $x = -3$ or $x = -5$
- (C) $x = 5$ or $x = 3$
- (D) $x = 3$ or $x = -3$
- (E) $x = 0$

Q8. Solve the equation $|3x - 2| = 7$.

- (A) $x = 1$ or $x = 3$
- (B) $x = 3$ or $x = 1$
- (C) $x = 3$ or $x = -3$
- (D) $x = -1$ or $x = 3$
- (E) $x = 0$

Open Ended Questions (FRQ)

Q9. Solve the equation $|2x - 5| = |x + 1|$ and check for extraneous solutions.

Q10. Solve the equation $|x - 2| = |3x + 1|$ and check for extraneous solutions.

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to split the equation into cases
- Simplify the resulting equations